Akoizo Original Practice - Form 3 - Answer key

Division B | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A missed reading should be entered as…

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

A. Missing with a reason

B. Zero automatically

C. The expected mean

D. A convenient nearby value

Correct: A

2. (2 points) The best final coherence check compares…

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

A. Only handwriting

B. Only the materials title

C. Question, variables, data, graph, and conclusion

D. Only page colors

Correct: C

3. (2 points) An inconvenient point should…

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

A. Always be deleted

B. Always be replaced by the mean

C. Be hidden in the graph

D. Be reviewed transparently using evidence

Correct: D

4. (2 points) Repeated readings on one plant always equal independent plants?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

A. No

B. Yes

C. Only at noon

D. Only with a calculator

Correct: A

5. (4 points) Explain “Mean” in your own words.

Unit: A clear Division B investigation report.
Worked-case data: 25 cm^2 towel: Absorbed masses: 2, 3, 4 g. 50 cm^2 towel: Absorbed masses: 5, 6, 7 g. Controls: Same towel brand, soak time, and draining method.
Separate investigation observations:
25 cm^2 towel: Absorbed masses: 2, 3, 4 g.
50 cm^2 towel: Absorbed masses: 5, 6, 7 g.
Controls: Same towel brand, soak time, and draining method.

answer: Sum divided by number of measurements. (4 points).

6. (4 points) Calculate mean and range for 8,10,12.

answer: Mean 10; range 4 response units. (4 points).

7. (4 points) Give three controlled conditions.

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

answer: Same object, surface, release method, travel distance, and timing method are examples. (4 points).

8. (4 points) Explain late stop timing direction.

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

answer: Stopping after the true endpoint increases recorded duration. (4 points).

9. (4 points) Why show individual trials?

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

answer: They reveal spread, unusual values, sample size, and patterns that a mean alone can hide. (4 points).

10. (4 points) What should be checked in the actual packet?

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

answer: Topic and material limits, timing, safety, required sections, and the applicable scoring expectations. (4 points).

11. (4 points) Write a useful follow-up question.

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

answer: Test whether the observed level-response trend persists with randomized order and stable temperature, using the same operational response and independent replication. (4 points).

12. (2 points) Define “Resolution” in the context of Experimental Design.

answer: Smallest distinguishable instrument increment under the stated reading method. (2 points).

13. (6 points) Solve this case and explain your reasoning: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

reasoning: Measure where time went: setup, resetting, recording, or coordination.
Simplify the apparatus or parallelize independent tasks while preserving definitions.
Reserve analysis time and test the revised workflow on a new prompt. (3 points).

conclusion: The improvement addresses the actual bottleneck rather than sacrificing evidence quality. (3 points).

14. (2 points) Define “Random error” in the context of Experimental Design.

answer: Unpredictable variation across observations. (2 points).

15. (2 points) Define “Raw data” in the context of Experimental Design.

answer: Original recorded measurements before summary processing. (2 points).

16. (2 points) Define “Operational definition” in the context of Experimental Design.

answer: Specific rule for how a variable is set or measured. (2 points).

17. (2 points) Define “Validity” in the context of Experimental Design.

answer: How well evidence supports the intended inference. (2 points).

18. (2 points) Define “Sensitivity analysis” in the context of Experimental Design.

answer: Checking how conclusions depend on a documented analytic choice. (2 points).